Simple Java Assignment: Creating and Using Classes

This assignment will walk you through the process of creating simple classes in Java, defining public variables within them, and accessing those variables from a main method. This is a foundational step in understanding how objects store data.

Part 1: Creating Classes and Reading Default Values

In this section, you will create three different classes. Each class will contain a few **public** instance variables. Public variables can be accessed and modified directly from other classes.

Your Task: Create Three Classes

1. **Create a new class** named Car.
    - Give it two public instance variables: make (a String) and year (an int).
2. **Create another class** named Dog.
    - Give it two public instance variables: name (a String) and age (an int).
3. **Create a third class** named Student.
    - Give it three public instance variables: firstName (a String), lastName (a String), and studentId (an int).

Your Task: The AssignmentTest Class

1. **Create a separate class** and name it AssignmentTest.
2. **Create a main method** within AssignmentTest.
3. **Inside the main method, create an instance of each of your three new classes.**
   Car myCar = new Car();
   Dog myDog = new Dog();
   Student myStudent = new Student();

4. **Print the values** of each public variable for all three objects. You will access the variables directly using the dot (.) operator. For example, myCar.make.
    - Notice the default values: null for String and 0 for int.

Example output for Part 1:
Car Make: null
Car Year: 0

Dog Name: null
Dog Age: 0

Student First Name: null
Student Last Name: null
Student ID: 0

Car.java (Part 1):

```java
// Car.java
public class Car {
    public String make;
    public int year;

    public Car() {
        // This is a default constructor. The variables will have their default
values.
        // For a String, this is null. For an int, it's 0.
    }
}
```

Dog.java (Part 1):

```java
// Dog.java
public class Dog {
    public String name;
    public int age;

    public Dog() {
        // This is a default constructor. The variables will have their default
values.
        // For a String, this is null. For an int, it's 0.
    }
}
```

Student.java (Part 1):

```java
// Student.java
public class Student {
    public String firstName;
    public String lastName;
    public int studentId;

    public Student() {
        // This is a default constructor. The variables will have their default
values.
    }
}
```

AssignmentTest.java (Part 1):

```java
// AssignmentTest.java
public class AssignmentTest {
    public static void main(String[] args) {
        System.out.println("--- Part 1: Default Values ---");

        // Creating instances of the classes using their default constructors.
        Car myCar = new Car();
        Dog myDog = new Dog();
        Student myStudent = new Student();

        // Printing the default values.
        System.out.println("Car Make: " + myCar.make);
        System.out.println("Car Year: " + myCar.year);
        System.out.println();

        System.out.println("Dog Name: " + myDog.name);
        System.out.println("Dog Age: " + myDog.age);
        System.out.println();

        System.out.println("Student First Name: " + myStudent.firstName);
        System.out.println("Student Last Name: " + myStudent.lastName);
        System.out.println("Student ID: " + myStudent.studentId);
    }
}
```

Part 2: Initializing Values

In this section, you will take the classes and objects you created in Part 1 and assign new values to their public variables.

Your Task: Modify the AssignmentTest Class

1. **Inside your three classes**, Car, Dog, and Student, **create a constructor that initializes all of the data in each class.**
   //goes inside class blueprint, underneath variables
   Car(int year, String make) {
   // ... SET VARIABLES TO INPUTS
   }
2. **After creating a constructor, create three instances of each class, using appropriate inputs** for all three objects to verify that they have been initialized.

Example output for Part 2:

Car1 Make: Honda
Car1 Year: 2022
Car2 Make: Lincoln
Car2 Year: 2000
Car2 Make:Lamborghini
Car2 Year: 2014

Dog1 Name: Sparky
Dog1 Age: 5
Dog2 Name: Bella
Dog2 Age: 8
Dog3 Name: Sparky
Dog3 Age: 5

Student1 First Name: Jane
Student1 Last Name: Doe
Student1 ID: 12345
Student2 First Name: John
Student2 Last Name: Matthew
Student2 ID: 45678

Student3 First Name: Mister
Student3 Last Name: Mann
Student3 ID: 91011

Car.java (Part 2):

```
//Car.java
public class Car {
    public String make;
    public int year;

    public Car() {
        // Default constructor
    }

    public Car(String carMake, int carYear) {
        // This is a parameterized constructor.
        // It initializes the make and year with the values passed in.
        make = carMake;
        year = carYear;
    }
}
```

Dog.java (Part 2):

```
// Dog.java
public class Dog {
    public String name;
    public int age;

    public Dog() {
        // Default constructor
    }

    public Dog(String dogName, int dogAge) {
        // This is a parameterized constructor.
        // It initializes the name and age with the values passed in.
        name = dogName;
        age = dogAge;
    }
}
```

Student.java (Part 2):

```java
// Student.java
public class Student {
    public String firstName;
    public String lastName;
    public int studentId;

    public Student() {
        // Default constructor
    }

    public Student(String studentFirstName, String studentLastName, int id) {
        // This is a parameterized constructor.
        // It initializes the student's information with the values passed in.
        firstName = studentFirstName;
        lastName = studentLastName;
        studentId = id;
    }
}
```

AssignmentTest.java (Part 2):

```java
// AssignmentTest.java
public class AssignmentTest {
    public static void main(String[] args) {
        System.out.println("--- Part 2: Parameterized Constructors ---");
        // Creating instances using the parameterized constructors.
        Car newCar = new Car("Toyota", 2024);
        Dog newDog = new Dog("Buddy", 5);
        Student newStudent = new Student("Jane", "Doe", 12345);

        // Printing the new values to confirm they were set correctly.
        System.out.println("New Car Make: " + newCar.make);
        System.out.println("New Car Year: " + newCar.year);
        System.out.println();

        System.out.println("New Dog Name: " + newDog.name);
        System.out.println("New Dog Age: " + newDog.age);
        System.out.println();
```

```
        System.out.println("New Student First Name: " + newStudent.firstName);
        System.out.println("New Student Last Name: " + newStudent.lastName);
        System.out.println("New Student ID: " + newStudent.studentId);
    }
}
```